

Project report

1. Introduction

Evolution is a complex process, in which certain beneficial traits usually get passed on and remain after multiple generations. These beneficial traits are usually created by random mutations and natural selection. For example, the long neck of giraffes is caused by mutations and natural selection favors the long neck traits causing the neck to become longer over multiple generations. As a result, I want to simulate mutations and natural selection in my project by building a simple model to see which trait is the most beneficial and how the proportions of breed types change over time. I also want to see how the model relates to emergence and see how the parameters affect the complexity of the system.

2. Model

The model works on a day loop which consists of a certain number of ticks determined by the days-length parameter. The model will start with a certain number of normal turtles controlled by the initial-number parameter. The model consists of one kind of turtle agent and one kind of grass patch on a 2D grid. The model also has a travel-distance parameter to control how far the turtle can move. Each patch of grass has a certain number of grass determined by the grass-amounts parameter. At the end of the day, all of the grass will regrow. The turtles can move, eat, reproduce, or die, and they need a certain amount of energy determined by the energy-needed parameter to survive to the next day. If the turtle does not eat enough energy at the end of the day, it will die. However, if the turtle eats more than the required energy at the end of the day, it will reproduce. The offspring has an 80% chance to stay the same breed as the parent, and a 20% chance to mutate to become one of other breeds. In a day, the turtles can move around and eat every two ticks. When the turtles on the same patch of grass attempt to eat, they will all fail to eat if the total amount of grass needed is larger than the total amount of grass left. The turtle agent has 5 different breed types which behave differently. The 5 breeds are normal, greedy, fast, fertile and fair:

- + Normal (Blue): These turtles eat 2 grass at a time and move every 2 ticks. Require the normal energy amount to survive.
- + Greedy (Red): These turtles eat 4 grass at a time and move every 2 ticks. Require the normal energy amount to survive.

- + Fast (Yellow): These turtles eat 2 grass at a time and move every tick. Requires double the energy amount to survive.
- + Fertile (Brown): These turtles eat 2 grass at a time and move every 2 ticks. Require the normal energy amount to survive. Give births to 2 turtles at a time.
- + Fair (Black): These turtles eat 3 grass at a time and move every 2 ticks. Require the normal energy amount to survive.

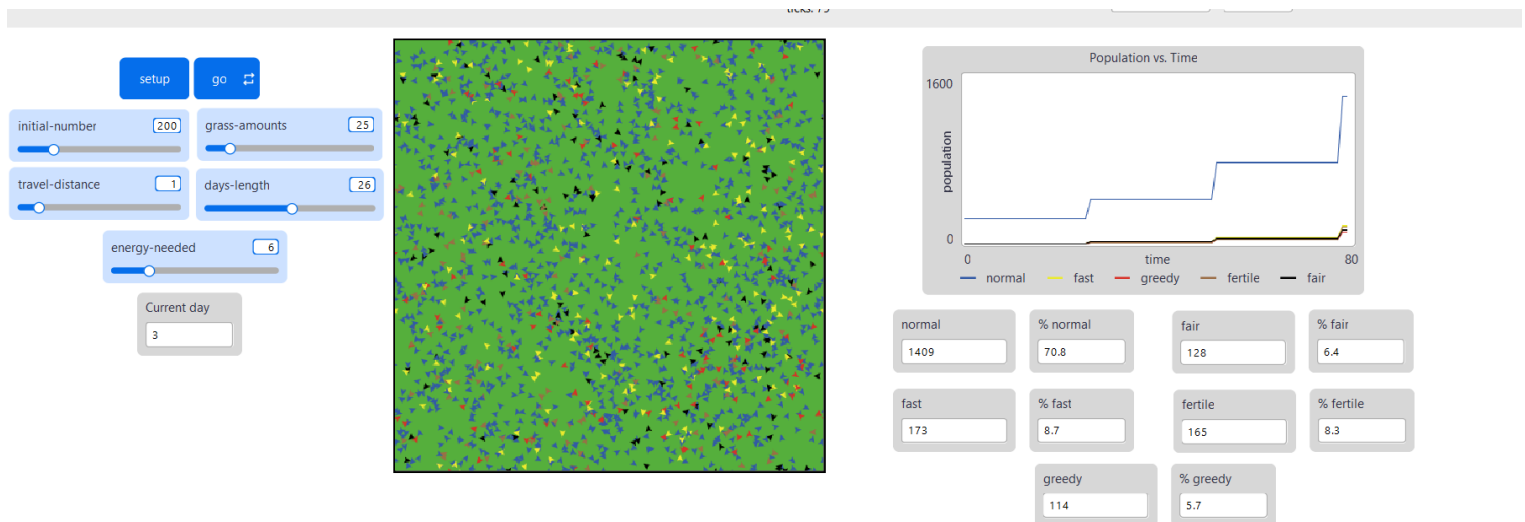


Figure 1. The model interface with multiple turtles and different breeds

3. Result

In the first few days, the normal turtles do not have much competition and begin to reproduce and increase their number rapidly. The other breeds also begin to appear and increase their number. After a few more days, the board becomes more crowded and it becomes more competitive for the turtles. Depending on the parameters, the proportion of breeds behaves differently. However, the number of normal turtles usually reach a peak before decreasing rapidly, as they can not compete with the greedy, fertile and fair breeds which get a beneficial trait to help them. For the fast breed, they can only reach up to around 10 - 15% of the proportion of turtles before decreasing and staying at quite low percentages. Although the fast breed can move and eat every tick which means they should have twice the chance of successfully eating than other breeds, the fact they need double the energy seems to be too much of a handicap preventing the fast breed from thriving.

For the other 3 breeds, they can overtake the normal breed but they can also dominate the other breeds depending on the parameters. When the energy

needed is extremely low at around 2, the fertile breed survives and reproduces a lot, leading them to dominate other breeds. When the energy is low at around 5, the fertile breed overtakes the normal breed very fast, but the fertile breed begins to decrease rapidly due to the increase of competition. On the other hand, the fair breed becomes the dominant breed. When the energy is 10 or higher, the greedy breed becomes the dominant breed as there is more competition where the greedy breed thrives the most.

The travel distance affects how far away a turtle can move, but it does not affect the distribution much. It seems to only change how much the proportions flutter around the equilibrium. The grass amounts and the day length parameters do affect the proportions but it is not as clear as the energy needed parameter. They determine how much the population of each turtle can grow in a day.

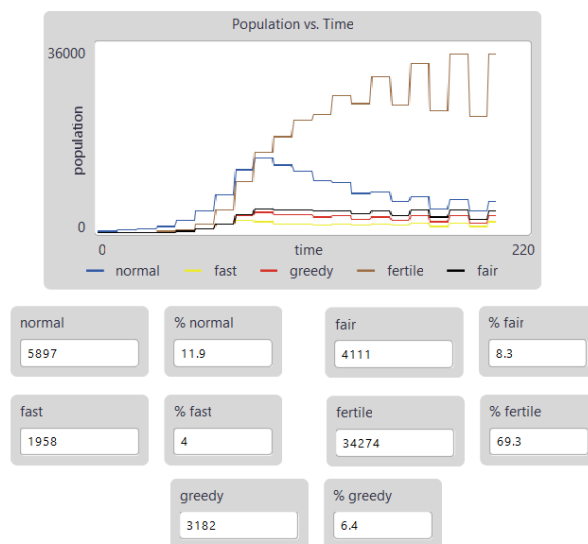


Figure 2. Population graphs at day 20 with initial-number = 200, travel-distance = 1, grass-amounts = 25, days-length = 10, energy needed = 2. The fertile breed is dominant.

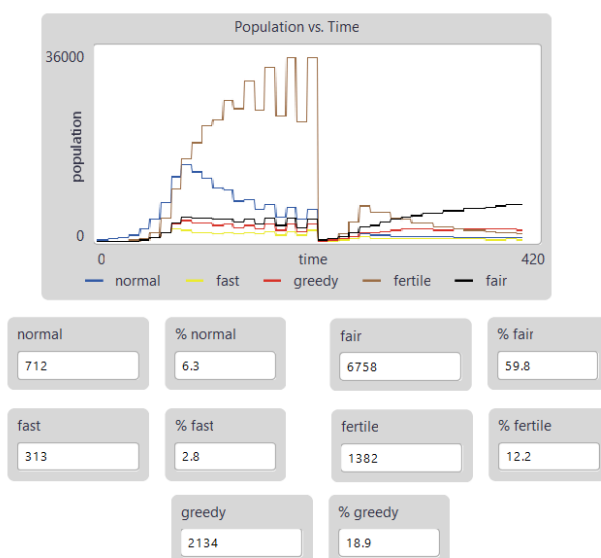


Figure 3. Population graphs at day 40 with grass-amounts = 25, days-length = 10, energy needed change to 5. Overpopulation and higher competition due to higher energy requirements cause mass death of every breed. The fair breed is leading.

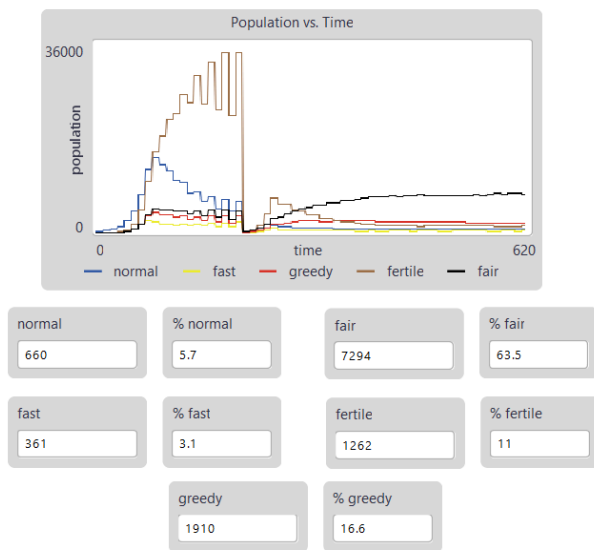


Figure 4. Population graphs at day 60 with grass-amounts = 25, days-length = 10, energy needed = 5. The fair breed is the dominant breed. Due to higher competition and lack of food, the population is lower.

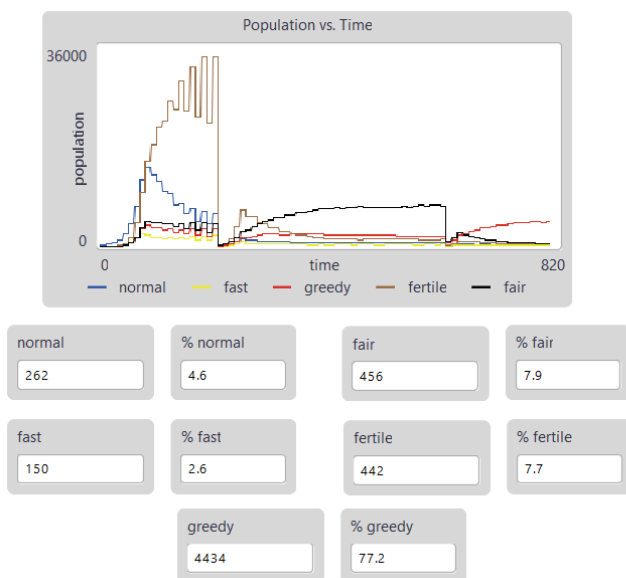


Figure 5. Population graphs at day 80 with grass-amounts = 25, days-length = 10, energy needed changes to 10. Due to higher competition and energy needed, it causes mass death again. The greedy breed becomes the dominant breed.

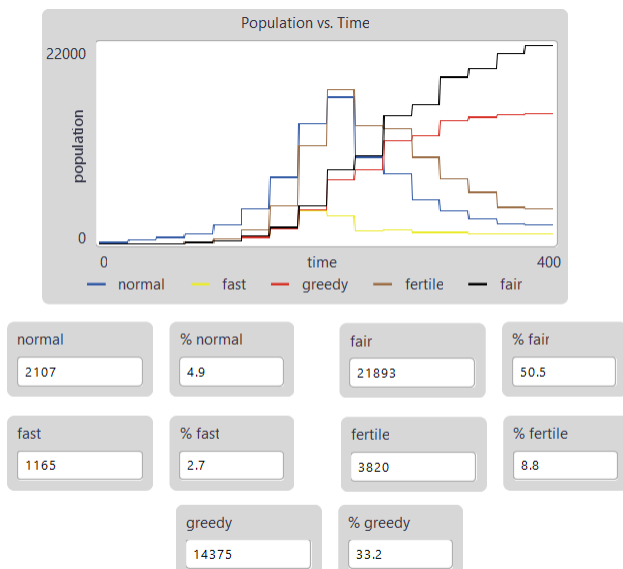


Figure 6. Population graphs at day 15 with initial-number = 200, travel-distance = 1, grass-amounts = 100, days-length = 25, energy needed = 5. The population. The fair breed becomes the dominant breed. The dominant breed reaches over 22000 turtles.

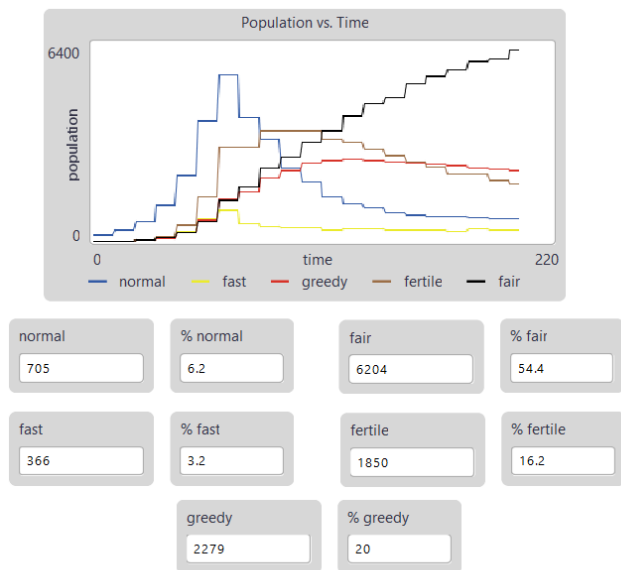


Figure 7. Population graphs at day 20 with initial-number = 200, travel-distance = 1, grass-amounts = 25, days-length = 10, energy needed = 5. The population. The fair breed becomes the dominant breed. The dominant breed only reaches just over 6400 turtles.

4. Conclusion

The system simulates the random mutations and natural selection successfully. The normal breed mutates and gets replaced by another breed with a beneficial trait that helps with the survivability of the breed. The different parameters and environments favor different breeds. In this system particularly, the energy needed is the biggest factor determining which breed is favored. This agent-based system also portrays an emergent population dynamic. Each turtle follows fairly simple instructions, but their interactions combining with the parameters leads to complex patterns at the population level like dominance shiftings and equilibriums. The parameters add some complexity to systems as the model and turtle population adapts to the new environment as the parameters changed. The system also displays strong negative feedback loops when the system stabilizes, as the population reduces when it gets too crowded, and the population increases again when it becomes less crowded.